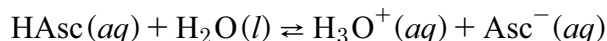
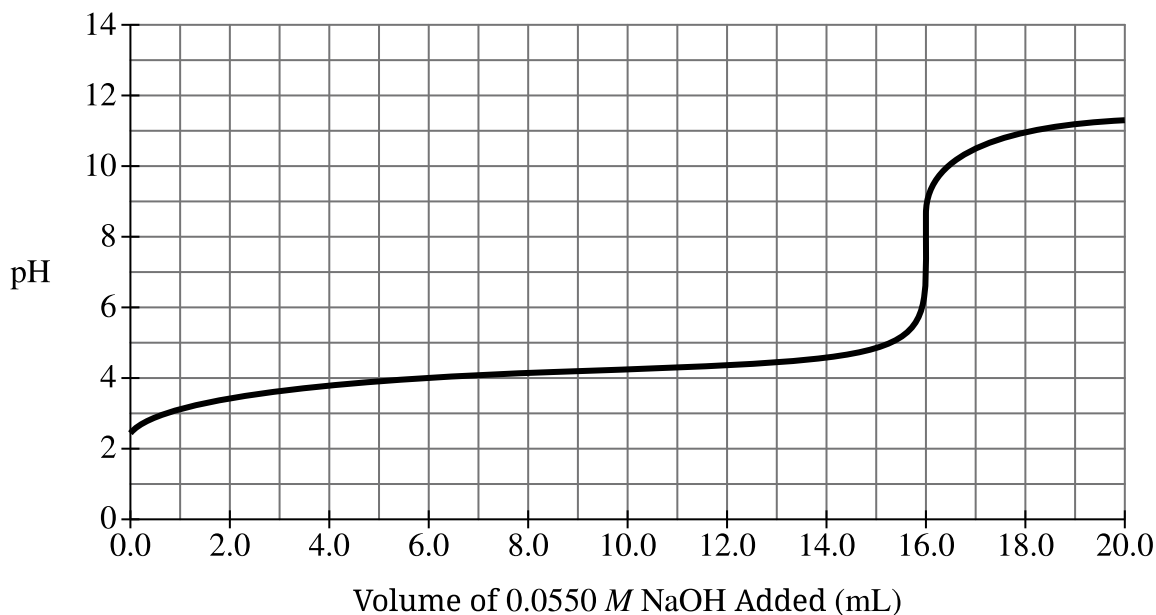


2. Answer the following questions about ascorbic acid (vitamin C).

- A. A student combusts a sample of ascorbic acid,  $C_xH_yO_z$ , to determine its chemical composition. The only products of the reaction are 0.2400 mol of  $CO_2$  and 2.883 g of  $H_2O$ .
- Calculate the number of moles of  $H_2O$  produced.
  - The mole ratio of carbon (C) to oxygen (O) is 1:1 in ascorbic acid. Based on this information and your answer to part A (i), determine the empirical formula of ascorbic acid.
- B. Ascorbic acid,  $HAsc(aq)$ , acts as a weak acid, as shown in the equation.



The following titration curve was produced when a 10.0 mL sample of  $HAsc(aq)$  was titrated using 0.0550 M  $NaOH(aq)$ .



- Calculate the molar concentration of the ascorbic acid solution.
- From the titration curve, determine the approximate  $pK_a$  of ascorbic acid.
- What is the value of the ratio  $\frac{[Asc^-]}{[HAsc]}$  when the pH of the solution is 4.7?

- C. Dehydroascorbic acid (DHAsc) can be produced by reacting ascorbic acid with the triiodide ion,  $\text{I}_3^-$ , as represented by the following equation.



The student runs three trials of the reaction with different initial concentrations of HAsc and  $\text{I}_3^-$ , producing the following data.

| Trial | [HAsc] (M) | $[\text{I}_3^-]$ (M) | Initial Rate of DHAsc Formation (M/s) |
|-------|------------|----------------------|---------------------------------------|
| 1     | 0.450      | 1.200                | $2.457 \times 10^{-4}$                |
| 2     | 0.450      | 0.600                | $1.229 \times 10^{-4}$                |
| 3     | 0.900      | 1.200                | $4.914 \times 10^{-4}$                |

- The rate law for the reaction is  $\text{rate} = k[\text{HAsc}][\text{I}_3^-]$ . Explain how the data in the table support the conclusion that the reaction is first order with respect to  $[\text{HAsc}]$ .
  - Calculate the value of the rate constant,  $k$ , for the reaction. Include units with your answer.
- D. The triiodide ion,  $\text{I}_3^-$ , is significantly more soluble in water than elemental iodine,  $\text{I}_2$ , is. Identify an intermolecular force between  $\text{I}_3^-$  and water that is **not** present between  $\text{I}_2$  and water, which could explain the difference in solubility. Lewis diagrams for  $\text{I}_2$  and  $\text{I}_3^-$  are provided.

